International AS · Edexcel · Chemistry

53 mins · 3 questions

Structured Questions

Organic Chemistry: Techniques & Spectra

Organic Chemistry Techniques / Interpreting Mass Spectra / Using Infrared Spectrometry

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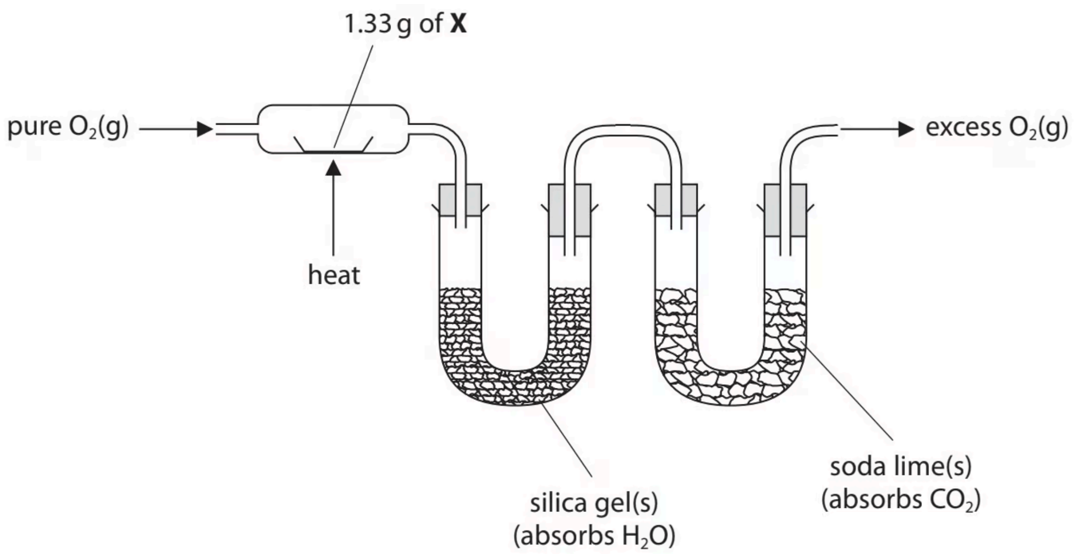
Total Marks **/53**

1 (a) This question is about two organic compounds, **X** and **Y.** Both are liquids which contain carbon, hydrogen and oxygen only.

The mass of hydrogen and of carbon present in 1.33 g of **X** were determined by passing its combustion products through the apparatus shown.



i) State the **measurements** that should be made.

(2)

ii) Give **two** reasons why pure O_2 (g), and **not** air, should be used.

(2)

iii) The experiment showed that 1.33 g of **X** contains 0.14 g of hydrogen and 0.63 g of carbon.

Calculate the empirical formula of **X**, using these data.

You **must** show your working.

(3)

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(7 marks)

(b) When phosphorus(V) chloride is added to **X**, steamy white fumes are seen.

State what can be deduced about compound **X** from this observation only.

....

(1 mark)

(c) Compound **X** is converted into compound **Y** when refluxed with **excess** sodium dichromate(VI) in sulfuric acid.

Compound **Y** is a liquid that is soluble in the reaction mixture.

Draw a **labelled** diagram of the apparatus that could be used to separate **Y** from the reaction mixture.

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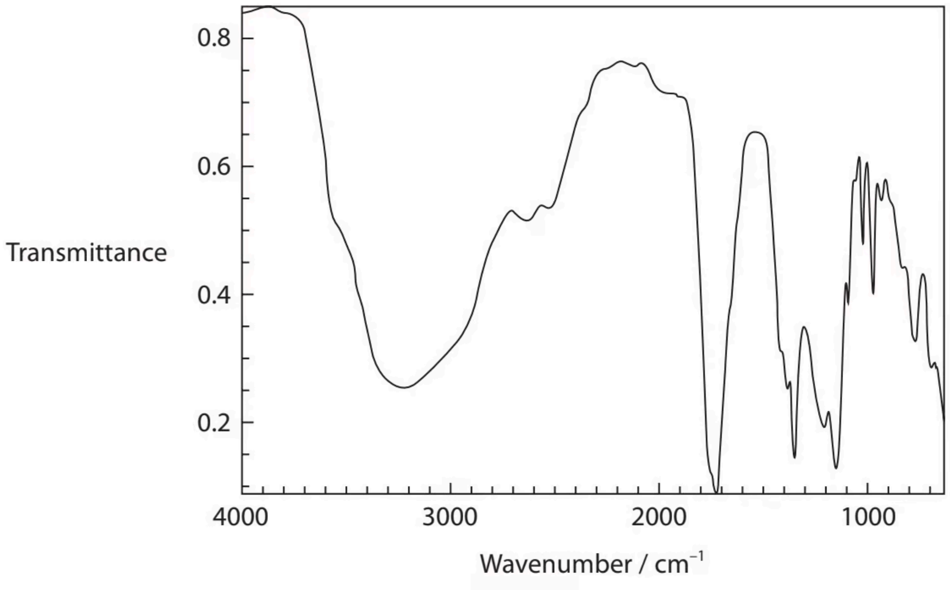
(3 marks)

(d) The infrared spectrum of **Y** is shown.



The table shows some infrared absorption data.

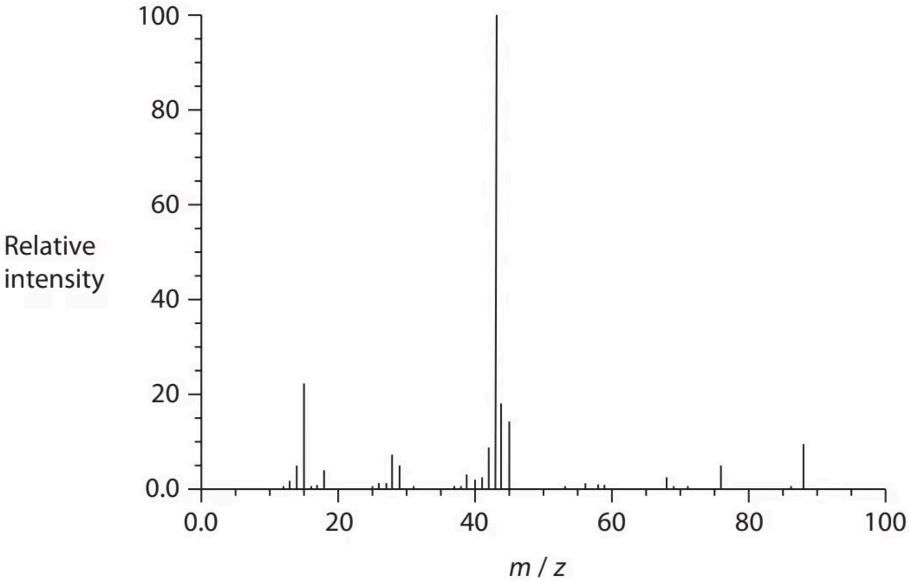
Bond	Wavenumber range / cm^{-1}
C—H (alkane)	2962–2853
O—H (alcohols and phenols)	3750–3200
O—H (carboxylic acids)	3300–2500
C = C (alkene)	1669–1645
C = O (aldehydes, ketones, carboxylic acids)	1740–1680

Explain how this spectrum shows that **Y** contains a carboxylic acid functional group, quoting data from the table.

....

(2 marks)

(e) The mass spectrum of **Y** is shown.



i) Show that the mass spectrum is consistent with **Y** having the molecular formula $C_3H_4O_3$.

(1)

ii) Suggest the structure of the ion causing the peak at *m / z* = 43 in the mass spectrum of **Y**.

(1)

(2 marks)

(f) Compound **X** contains one type of functional group.

Compound **Y** contains two different functional groups.

Use the information in the question to deduce the structures of **X** and **Y**.

Compound **X**

Compound **Y**

....

....

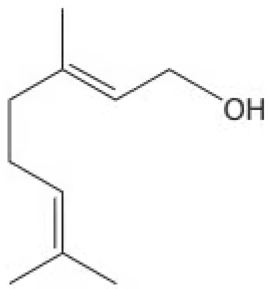
(2 marks)

2 (a) Geraniol is used in perfumes and can be extracted from many plants.

Data on geraniol are shown.

Solubility in water	Melting temperature/°C	Boiling temperature/ °C	Density / g cm^{-3}
insoluble	−15	230	0.889

The structure of geraniol is shown.



Geraniol has **two** different types of functional group.

Name the functional groups, giving a chemical test and its positive result to show the presence of each group.

First functional group:

Second functional group:

....

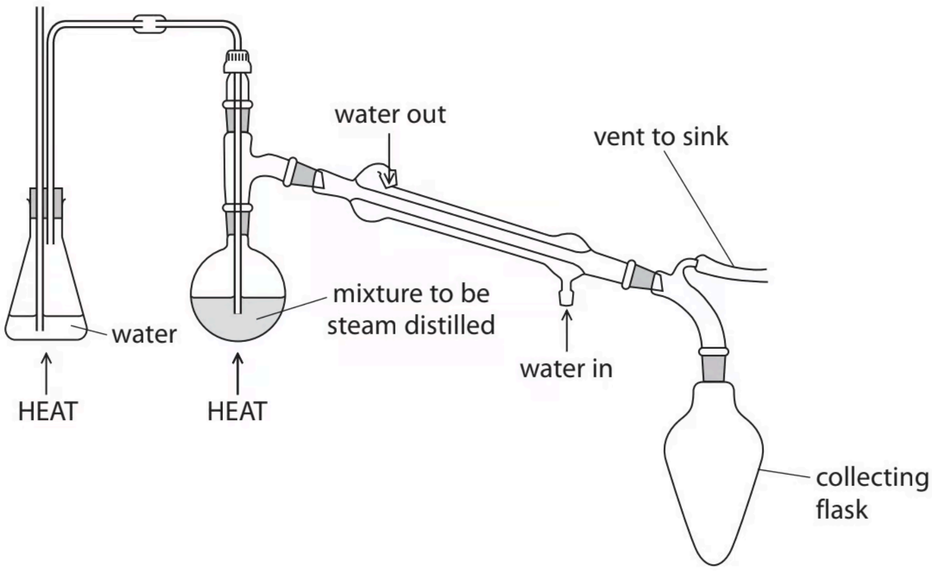
....

....

....

(4 marks)

(b) Geraniol is extracted by steam distillation.



The steam distillation product is geraniol and water. The water may contain dissolved impurities which have similar boiling temperatures to geraniol.

The contents of the collecting flask are transferred to a piece of apparatus used to separate the geraniol from the water layer.

Draw a labelled diagram of this apparatus and its contents.

....

....

....

(3 marks)

(c) The geraniol will still contain small quantities of water.

Describe how to produce a sample of pure, dry geraniol using a named drying agent.

....

....

....

(3 marks)

(d) The hazard labels for pure geraniol are shown.

i) Complete the table to identify the hazards indicated by the symbols.

(2)

ii) State **one** precaution, other than wearing safety spectacles and a laboratory coat, that should be taken when using pure geraniol to reduce the risk associated with the hazard symbol shown.

(1)

(3 marks)

(e) State the appearance of the flame when geraniol is ignited.

(1 mark)

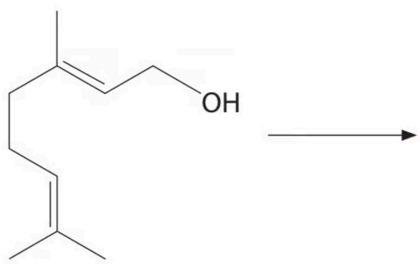
(f) Geraniol reacts with **excess** hydrogen gas.

i) State the essential condition required for this reaction.

(1)

ii) Draw the **skeletal** formula of the product of this reaction.

(1)

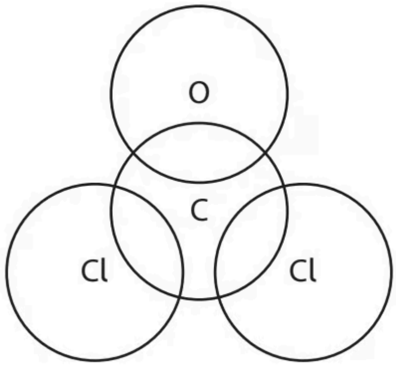


geraniol

(2 marks)

3 (a) Phosgene ($COCl_2$) is a colourless gas used in the pharmaceutical industry. Phosgene has a boiling temperature of 8 °C and is extremely toxic.

Complete the dot-and-cross diagram to show the bonding in phosgene.



(2 marks)

(b) Phosgene can be formed from carbon monoxide and chlorine, using a catalyst of activated carbon.

$CO\,(g) + Cl_2\,(g) \rightleftharpoons COCl_2\,(g)$ $\Delta_r H = -107.6\ kJ\ mol^{-1}$

i) State and explain how the reaction conditions could be changed to maximise the **equilibrium** yield of phosgene in this reaction.

(4)

ii) The standard enthalpy change of formation for phosgene is $\Delta_f H = -220.1\ kJ\ mol^{-1}$.

Complete the Hess cycle and determine the standard enthalpy change of formation for carbon monoxide. Use the data from (b)(i).

Include state symbols in your cycle.

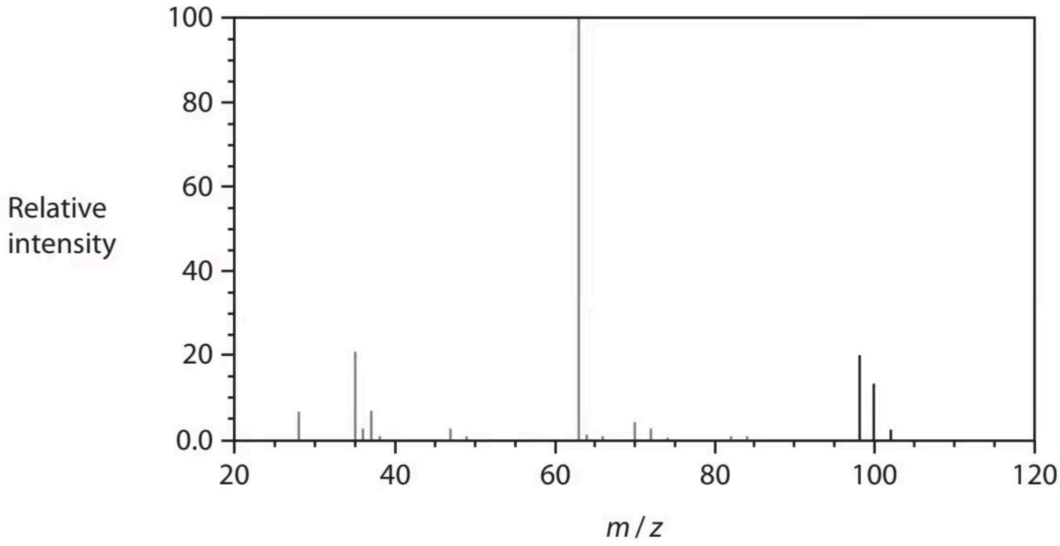
(4)

$$CO(g) + Cl_2\,(g) \rightarrow COCl_2\,(g)$$

(8 marks)

(c) The mass spectrum of a sample of phosgene is shown.

The peak at *m / z* = 65 has been omitted.



i) Give the reason for the **ratio** of peak heights at *m / z* values of 102, 100 and 98

(2)

ii) Suggest an identity for the peak at *m / z* = 63.

(1)

iii) The peak at *m* / *z* = 65 has been omitted.

Draw **on the mass spectrum** the peak at *m* / *z* = 65, showing its relative intensity.

(1)

(4 marks)

(d) Use your Data Booklet to suggest the wavenumber of a strong absorbance you would expect to see in the infrared spectrum for phosgene. Justify your answer.

(2 marks)

(e) In UV light, trichloromethane ($CHCl_3$, boiling temperature 61 °C) reacts with oxygen to form phosgene and hydrogen chloride.

i) Write an equation for this reaction. State symbols are not required.

(1)

ii) In a closed bottle, the rate of this reaction decreases with time.

Give a reason for this.

(1)

iii) Suggest a precaution that should be taken when opening a bottle of trichloromethane.

(1)

iv) Trichloromethane can be used as an anaesthetic.

Suggest whether an old bottle of trichloromethane can still be used for medical treatment, giving a reason for your answer.

(1)

(4 marks)